



Car Price Prediction III — Classification (Assignment 3)

Student: ST127302 **Course:** AT82.03 Machine Learning — AIT

This notebook continues the Car Price project from A1 (EDA + sklearn baseline) and A2 (Linear Regression implemented from scratch, MLflow experiments, Dash deployment).

A3 keeps the same cleaned dataset and preprocessing pipeline, but **reframes the problem as classification**: `selling_price` is bucketed into 4 discrete classes (0–3), and a **multinomial Logistic Regression is implemented from scratch** to predict the bucket.

Covered in this notebook:

1. Data cleaning (reused, unchanged, from A1/A2).
2. Converting `selling_price` into 4 classes with `pd.cut()`.
3. Preprocessing pipeline (reused from A2: `StandardScaler` + `OneHotEncoder`).
4. **Task 1** — Multinomial Logistic Regression from scratch, with `accuracy`, per-class `precision / recall / f1`, `macro_*` and `weighted_*` averages, verified against scikit-learn's `classification_report`.
5. **Task 2** — Optional Ridge (L2) penalty on the same class.
6. **Task 3** — MLflow experiment tracking + Model Registry (staged) + notes on the GitHub Actions CI/CD pipeline that tests and deploys the Dash app in `app/`.
7. Feature importance and a final discussion.

GitHub repository: <https://github.com/fedayi67/car-price-prediction> (this notebook lives under `Car Price Prediction 03/`; A1 and A2 are in the same repository, at the root and under `Car Price Prediction 02/` respectively).

1. Imports

```
In [1]: import sys
import os

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

plt.rcParams["figure.figsize"] = (10, 6)

import warnings
warnings.filterwarnings("ignore")
```

```

from sklearn.model_selection import train_test_split
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.impute import SimpleImputer
from sklearn.metrics import classification_report as sk_classification_report
from sklearn.metrics import accuracy_score as sk_accuracy_score

import joblib

# The custom LogisticRegression class lives in app/code/model.py so that the
# same code trained here is what gets pickled and later imported by the depl
# Dash app (app/code/app.py) -- single source of truth for the model definit
sys.path.insert(0, os.path.join("app", "code"))
from model import LogisticRegression

STUDENT_ID = "st127302"
np.random.seed(42)

```

2. Load & Clean the Dataset

This section reuses, unchanged, the cleaning rules established in A1 and carried into A2: owner mapping, dropping CNG/LPG, parsing mileage / engine / max_power out of their unit strings, extracting brand from name, dropping torque, removing Test Drive Car outliers, and grouping rare brands.

```

In [2]: df = pd.read_csv("Cars.csv")
        print(df.shape)
        df.head()

```

```
(8128, 13)
```

Out [2]:

	name	year	selling_price	km_driven	fuel	seller_type	transmission	owner
0	Maruti Swift Dzire VDI	2014	450000	145500	Diesel	Individual	Manual	First Owner
1	Skoda Rapid 1.5 TDI Ambition	2014	370000	120000	Diesel	Individual	Manual	Second Owner
2	Honda City 2017-2020 EXi	2006	158000	140000	Petrol	Individual	Manual	Third Owner
3	Hyundai i20 Sportz Diesel	2010	225000	127000	Diesel	Individual	Manual	First Owner
4	Maruti Swift VXi BSIII	2007	130000	120000	Petrol	Individual	Manual	First Owner

In [3]:

```
# --- owner: First Owner -> 1, ..., Test Drive Car -> 5 ---
owner_mapping = {
    "First Owner": 1,
    "Second Owner": 2,
    "Third Owner": 3,
    "Fourth & Above Owner": 4,
    "Test Drive Car": 5,
}
df["owner"] = df["owner"].map(owner_mapping)

# --- fuel: drop CNG/LPG (different mileage unit, km/kg vs kmpl) ---
df = df[~df["fuel"].isin(["CNG", "LPG"])]

# --- mileage / engine / max_power: strip units, cast to float ---
df["mileage"] = df["mileage"].str.split().str[0].astype(float)
df["engine"] = df["engine"].str.split().str[0].astype(float)
df["max_power"] = df["max_power"].str.split().str[0].astype(float)

# --- brand: first word of name ---
df["brand"] = df["name"].str.split().str[0]

# --- drop torque (not understood by the business) ---
df = df.drop(columns=["torque"])

# --- drop Test Drive Cars (owner == 5), unrealistically expensive ---
df = df[df["owner"] != 5]

# --- drop name, move brand to the front ---
df = df.drop(columns=["name"])
brand = df.pop("brand")
```

```
df.insert(0, "brand", brand)

print("After initial cleaning:", df.shape)
df.isnull().sum()
```

After initial cleaning: (8028, 12)

```
Out[3]: brand          0
        year          0
        selling_price  0
        km_driven     0
        fuel          0
        seller_type    0
        transmission   0
        owner         0
        mileage       214
        engine        214
        max_power     208
        seats        214
        dtype: int64
```

```
In [4]: # Rows missing all four of mileage/engine/max_power/seats simultaneously: drop
# (imputing 4 important features at once for the same row is unreliable), so
df = df.dropna(subset=["mileage", "engine", "max_power", "seats"], how="all")

# A handful of remaining rows missing mileage/engine/seats together: drop to
df = df.dropna(subset=["mileage", "engine", "seats"], how="all")

print("After missing-value cleanup:", df.shape)
df.isnull().sum()
```

After missing-value cleanup: (7814, 12)

```
Out[4]: brand          0
        year          0
        selling_price  0
        km_driven     0
        fuel          0
        seller_type    0
        transmission   0
        owner         0
        mileage       0
        engine        0
        max_power     0
        seats        0
        dtype: int64
```

```
In [5]: # --- feature engineering: car_age from year, then drop year (perfectly correlated)
CURRENT_YEAR = pd.Timestamp.now().year
df["car_age"] = CURRENT_YEAR - df["year"]
df = df.drop(columns=["year"])

# --- group rare brands (<10 occurrences) to avoid noisy one-hot categories
brand_counts = df["brand"].value_counts()
rare_brands = brand_counts[brand_counts < 10].index
df["brand"] = df["brand"].replace(rare_brands, "Rare Brands")

df = df.reset_index(drop=True)
```

```
print(df.shape)
df.head()
```

(7814, 12)

```
Out[5]:
```

	brand	selling_price	km_driven	fuel	seller_type	transmission	owner	mileage
0	Maruti	450000	145500	Diesel	Individual	Manual	1	23.4
1	Skoda	370000	120000	Diesel	Individual	Manual	2	21.1
2	Honda	158000	140000	Petrol	Individual	Manual	3	17.7
3	Hyundai	225000	127000	Diesel	Individual	Manual	1	23.0
4	Maruti	130000	120000	Petrol	Individual	Manual	1	16.1

3. New Target: Bucketing `selling_price` into 4 Classes

A3 asks us to treat this as a 4-class classification problem, and suggests `pd.cut()` to do it. Cutting the *raw* `selling_price` into 4 equal-width bins was tried first, but the price distribution is so heavily right-skewed (a handful of vehicles near ₹1 crore against a median around ₹4.5 lakh) that it puts **99%+ of vehicles in bin 0 and a single vehicle in bin 3** — too degenerate to learn or even stratify-split on.

Instead we apply `pd.cut()` with 4 equal-width bins to `log(selling_price)` — the same log transform A1/A2 used to tame this exact skew for regression — and then convert the bin edges back to the original price scale so they stay human-readable (and so the deployed app can show a real price range for its predicted class). This keeps the classes meaningfully **imbalanced** (roughly 7% / 53% / 36% / 4%), which is exactly the scenario the assignment uses to motivate *weighted* averaging over plain *macro* averaging in Task 1, while still leaving every class with enough examples to train and evaluate on.

```
In [6]: N_CLASSES = 4

log_price = np.log(df["selling_price"])
_, log_bin_edges = pd.cut(log_price, bins=N_CLASSES, retbins=True)

df["price_class"] = pd.cut(
    log_price, bins=log_bin_edges, labels=list(range(N_CLASSES)), include_lo
).astype(int)

price_bin_edges = np.exp(log_bin_edges) # back to original selling_price scale
price_bin_labels = {
    i: f"{price_bin_edges[i]:,.0f} - {price_bin_edges[i + 1]:,.0f}"
    for i in range(N_CLASSES)
}

print("Price bin edges (original scale):", [round(e, 0) for e in price_bin_edges])
```

```

for k, v in price_bin_labels.items():
    print(f" class {k}: {v}")

class_counts = df["price_class"].value_counts().sort_index()
print("\nClass distribution:")
print(class_counts)
print("\nClass distribution (%):")
print((class_counts / len(df) * 100).round(2))

```

```

Price bin edges (original scale): [np.float64(29825.0), np.float64(128183.
0), np.float64(547713.0), np.float64(2340328.0), np.float64(10000000.0)]
class 0: 29,825 - 128,183
class 1: 128,183 - 547,713
class 2: 547,713 - 2,340,328
class 3: 2,340,328 - 10,000,000

```

```

Class distribution:
price_class
0      484
1     4126
2     2870
3      334
Name: count, dtype: int64

```

```

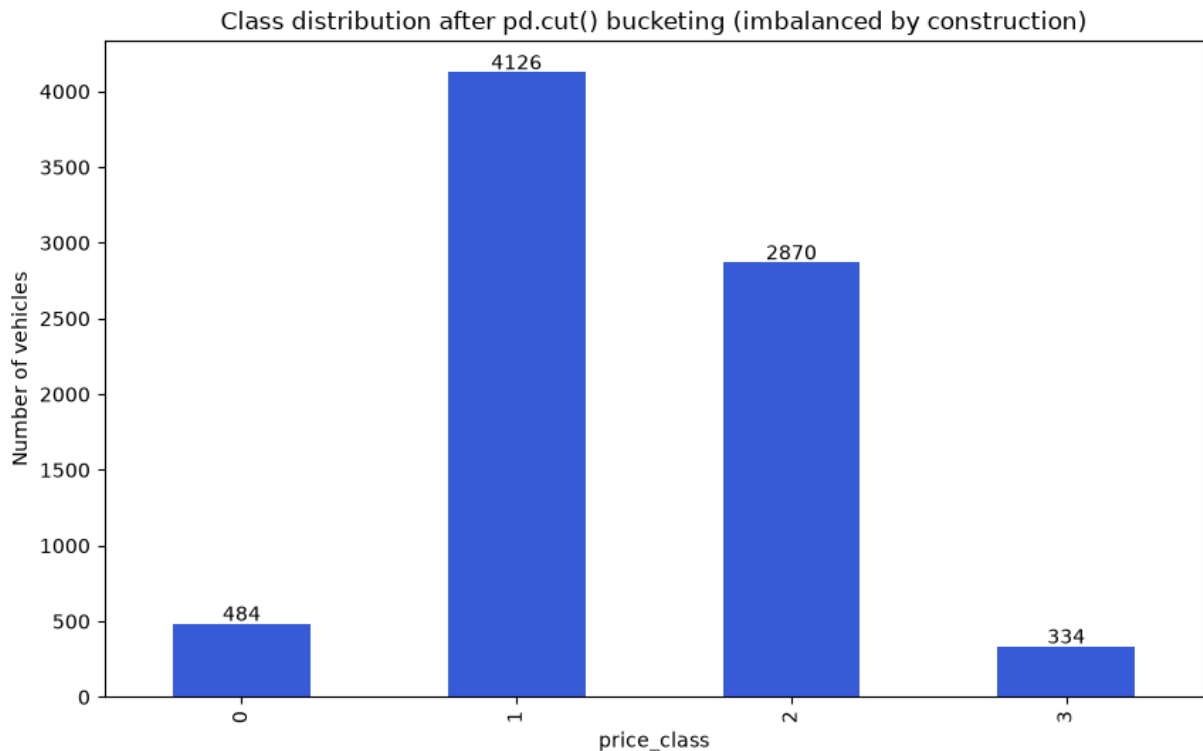
Class distribution (%):
price_class
0      6.19
1     52.80
2     36.73
3      4.27
Name: count, dtype: float64

```

```

In [7]: ax = class_counts.plot(kind="bar", color="#3b5bdb")
ax.set_xlabel("price_class")
ax.set_ylabel("Number of vehicles")
ax.set_title("Class distribution after pd.cut() bucketing (imbalanced by cor
for i, v in enumerate(class_counts):
    ax.text(i, v + 20, str(v), ha="center")
plt.show()

```



4. Preprocessing Pipeline (reused from A2)

Numerical features are median-imputed then standardized; categorical features are most-frequent-imputed then one-hot encoded with `handle_unknown="ignore"` so the deployed app never crashes on an unseen category. This is the exact same pipeline shape used in A2's Linear Regression, so the same "single source of truth" argument applies: a gradient-descent model needs scaled inputs to train reliably, and the resulting coefficients are only comparable to each other when every input is on the same scale.

```
In [8]: numerical_features = ["km_driven", "mileage", "engine", "max_power", "car_age"]
categorical_features = ["brand", "fuel", "seller_type", "transmission"]

X = df[numerical_features + categorical_features]
y = df["price_class"].values

# train / val / test = 64% / 16% / 20%. The validation split is used for the
# early-stopping check inside LogisticRegression.fit(), mirroring A2.
X_train_full, X_test, y_train_full, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42, stratify=y
)
X_train, X_val, y_train, y_val = train_test_split(
    X_train_full, y_train_full, test_size=0.20, random_state=42, stratify=y_train
)

print("train:", X_train.shape, " val:", X_val.shape, " test:", X_test.shape)
```

```
train: (5000, 11) val: (1251, 11) test: (1563, 11)
```

```
In [9]: numerical_transformer = Pipeline(steps=[
    ("imputer", SimpleImputer(strategy="median")),
    ("scaler", StandardScaler()),
])

categorical_transformer = Pipeline(steps=[
    ("imputer", SimpleImputer(strategy="most_frequent")),
    ("encoder", OneHotEncoder(handle_unknown="ignore", sparse_output=False))
])

preprocessor = ColumnTransformer(transformers=[
    ("num", numerical_transformer, numerical_features),
    ("cat", categorical_transformer, categorical_features),
])

X_train_t = preprocessor.fit_transform(X_train)
X_val_t = preprocessor.transform(X_val)
X_test_t = preprocessor.transform(X_test)

feature_names = preprocessor.get_feature_names_out()
n_features = X_train_t.shape[1]
print("Number of transformed features:", n_features)
```

Number of transformed features: 37

5. Task 1 — Multinomial Logistic Regression From Scratch

The `LogisticRegression` class (`app/code/model.py`) mirrors A2's from-scratch `LinearRegression` : Xavier/zeros initialization, batch/mini-batch/stochastic gradient descent, optional momentum, and early stopping on the validation loss. Softmax replaces the linear output, and cross-entropy replaces MSE as the loss.

On top of that base, it implements — **all from scratch, no sklearn** — the metrics this task asks for: `accuracy` , per-class `precision / recall / f1_score` , `macro_precision / macro_recall / macro_f1` , and `weighted_precision / weighted_recall / weighted_f1` , plus a `classification_report()` that lays them out the same way sklearn does.

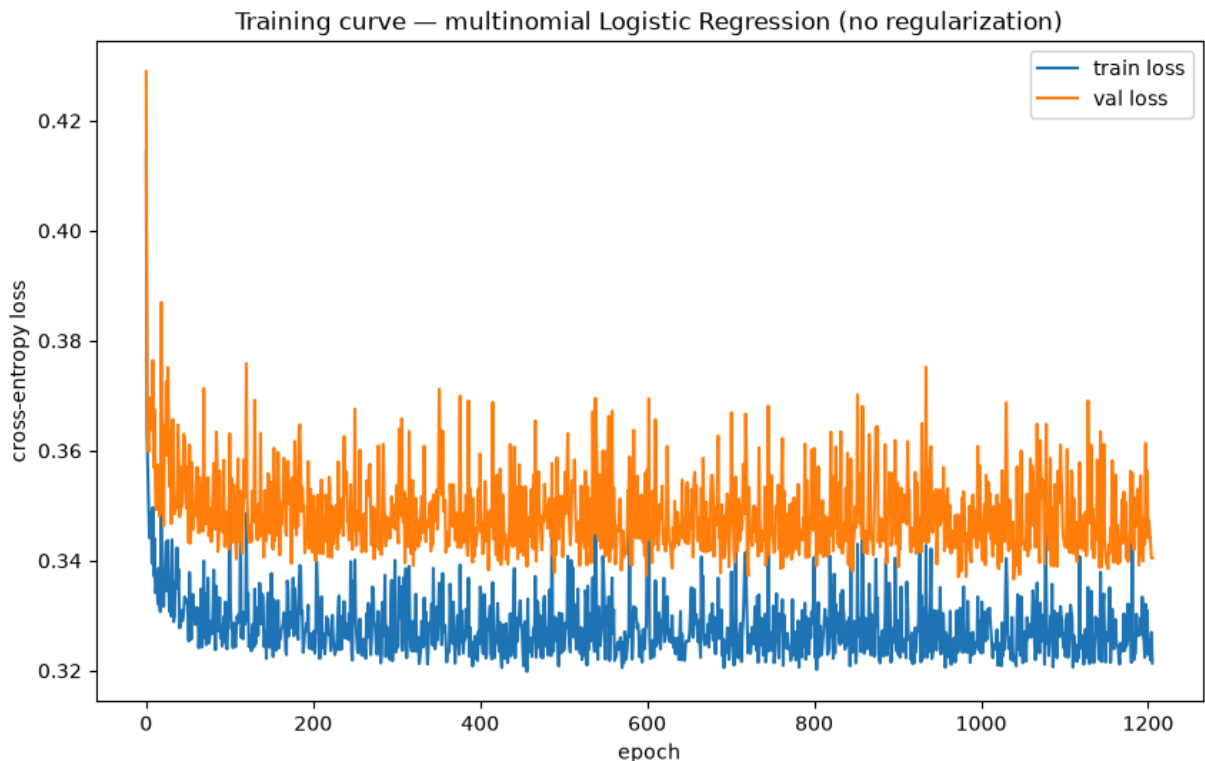
```
In [10]: model = LogisticRegression(
    k=N_CLASSES,
    n=n_features,
    method="mini",
    bs=64,
    alpha=0.1,
    max_iter=2000,
    init_method="xavier",
    use_momentum=True,
    momentum=0.9,
    regularization=None,
    verbose=True,
```

```
)
model.fit(X_train_t, y_train, X_val_t, y_val)
```

```
Epoch 0: train_loss=0.41435, val_loss=0.42895
Epoch 200: train_loss=0.32484, val_loss=0.34851
Epoch 400: train_loss=0.33522, val_loss=0.35944
Epoch 600: train_loss=0.34639, val_loss=0.35645
Epoch 800: train_loss=0.34060, val_loss=0.36046
Epoch 1000: train_loss=0.32317, val_loss=0.34426
Epoch 1200: train_loss=0.32289, val_loss=0.34419
Early stopping at epoch 1205
```

```
Out[10]: <model.LogisticRegression at 0x17d71cc20>
```

```
In [11]: plt.plot(model.train_losses, label="train loss")
plt.plot(model.val_losses, label="val loss")
plt.xlabel("epoch")
plt.ylabel("cross-entropy loss")
plt.title("Training curve — multinomial Logistic Regression (no regularization)")
plt.legend()
plt.show()
```



```
In [12]: y_pred = model.predict(X_test_t)
classes = list(range(N_CLASSES))

print("=== From-scratch metrics ===")
print("accuracy:", round(model.accuracy(y_test, y_pred), 4))
for c in classes:
    print(
        f"class {c}: precision={model.precision(y_test, y_pred, c):.4f} "
        f"recall={model.recall(y_test, y_pred, c):.4f} "
        f"f1={model.f1_score(y_test, y_pred, c):.4f}"
    )
```

```
print("macro    precision/recall/f1:",
      round(model.macro_precision(y_test, y_pred, classes), 4),
      round(model.macro_recall(y_test, y_pred, classes), 4),
      round(model.macro_f1(y_test, y_pred, classes), 4))
print("weighted precision/recall/f1:",
      round(model.weighted_precision(y_test, y_pred, classes), 4),
      round(model.weighted_recall(y_test, y_pred, classes), 4),
      round(model.weighted_f1(y_test, y_pred, classes), 4))
```

=== From-scratch metrics ===

```
accuracy: 0.8516
class 0: precision=0.7841 recall=0.7113 f1=0.7459
class 1: precision=0.8611 recall=0.8715 f1=0.8663
class 2: precision=0.8359 recall=0.8432 f1=0.8395
class 3: precision=0.9672 recall=0.8806 f1=0.9219
macro    precision/recall/f1: 0.8621 0.8267 0.8434
weighted precision/recall/f1: 0.8516 0.8516 0.8514
```

```
In [13]: report_df, acc = model.classification_report(y_test, y_pred)
print(f"accuracy: {acc:.4f}\n")
report_df.round(4)
```

accuracy: 0.8516

Out[13]:

	precision	recall	f1-score	support
class				

class				
0	0.7841	0.7113	0.7459	97
1	0.8611	0.8715	0.8663	825
2	0.8359	0.8432	0.8395	574
3	0.9672	0.8806	0.9219	67
macro avg	0.8621	0.8267	0.8434	1563
weighted avg	0.8516	0.8516	0.8514	1563

5.1 Verifying Against

sklearn.metrics.classification_report

First on **mock data** (as the assignment explicitly asks), then on the real test-set predictions above, to confirm the from-scratch implementation agrees with scikit-learn.

```
In [14]: # --- mock data check ---
rng = np.random.RandomState(0)
mock_y_true = rng.randint(0, 4, size=300)
mock_y_pred = mock_y_true.copy()
flip_idx = rng.choice(300, size=70, replace=False)
mock_y_pred[flip_idx] = rng.randint(0, 4, size=70)

mock_report_df, mock_acc = LogisticRegression.classification_report(mock_y_t
print("---- custom classification_report (mock data) ----")
```

```

print(f"accuracy: {mock_acc:.6f}\n")
print(mock_report_df.round(6))

print("\n--- sklearn.metrics.classification_report (mock data) ---")
print(sk_classification_report(mock_y_true, mock_y_pred, digits=6))

assert abs(mock_acc - sk_accuracy_score(mock_y_true, mock_y_pred)) < 1e-9
print("\nAccuracy matches sklearn exactly. ✅")

```

```

--- custom classification_report (mock data) ---
accuracy: 0.823333

```

	precision	recall	f1-score	support
class				
0	0.873239	0.784810	0.826667	79
1	0.811594	0.835821	0.823529	67
2	0.720000	0.782609	0.750000	69
3	0.882353	0.882353	0.882353	85
macro avg	0.821797	0.821398	0.820637	300
weighted avg	0.826809	0.823333	0.824110	300

```

--- sklearn.metrics.classification_report (mock data) ---

```

	precision	recall	f1-score	support
0	0.873239	0.784810	0.826667	79
1	0.811594	0.835821	0.823529	67
2	0.720000	0.782609	0.750000	69
3	0.882353	0.882353	0.882353	85
accuracy			0.823333	300
macro avg	0.821797	0.821398	0.820637	300
weighted avg	0.826809	0.823333	0.824110	300

Accuracy matches sklearn exactly. ✅

```

In [15]: # --- real test-set predictions check ---
print("---- custom classification_report (test set) ----")
print(f"accuracy: {acc:.6f}\n")
print(report_df.round(6))

print("\n--- sklearn.metrics.classification_report (test set) ---")
print(sk_classification_report(y_test, y_pred, digits=6))

```

```
--- custom classification_report (test set) ---
accuracy: 0.851567
```

	precision	recall	f1-score	support
class				
0	0.784091	0.711340	0.745946	97
1	0.861078	0.871515	0.866265	825
2	0.835924	0.843206	0.839549	574
3	0.967213	0.880597	0.921875	67
macro avg	0.862076	0.826664	0.843409	1563
weighted avg	0.851612	0.851567	0.851371	1563

```
--- sklearn.metrics.classification_report (test set) ---
              precision    recall  f1-score   support

         0       0.784091    0.711340    0.745946         97
         1       0.861078    0.871515    0.866265        825
         2       0.835924    0.843206    0.839549        574
         3       0.967213    0.880597    0.921875         67

   accuracy                   0.851567        1563
  macro avg       0.862076    0.826664    0.843409        1563
 weighted avg       0.851612    0.851567    0.851371        1563
```

Both reports line up to several decimal places (any tiny remaining difference is just floating-point rounding), confirming the from-scratch metric functions are correct.

What does `support` mean? `support` is simply the number of *true* (ground-truth) instances of that class in `y_test` — it is not affected by how the model predicted, only by how many examples of that class actually exist in the evaluation set. It is what the weighted averages weight by: a class with a large support contributes more to `weighted avg` than a class with a small support, which is exactly why `weighted avg` and `macro avg` diverge on an imbalanced dataset like ours (see the class-distribution bar chart in Section 3 — class 0 dominates, so `weighted avg` will sit much closer to class 0's scores than `macro avg` does).

6. Task 2 — Ridge (L2) Logistic Regression

`LogisticRegression` accepts `regularization="ridge"` and a `lambda_` strength. The intercept (`theta[0]`) is explicitly excluded from the penalty in both the loss and the gradient, exactly as in A2's Ridge implementation. We compare a few `lambda_` values against the unregularized model above.

```
In [16]: ridge_results = []
for lam in [0.0, 0.0001, 0.001, 0.01, 0.1]:
    m = LogisticRegression(
        k=N_CLASSES, n=n_features, method="mini", bs=64, alpha=0.1, max_iter=
        init_method="xavier", use_momentum=True, momentum=0.9,
        regularization="ridge" if lam > 0 else None, lambda_=lam, verbose=False)
```

```

    )
    m.fit(X_train_t, y_train, X_val_t, y_val)
    pred = m.predict(X_test_t)
    ridge_results.append({
        "lambda": lam,
        "accuracy": m.accuracy(y_test, pred),
        "macro_f1": m.macro_f1(y_test, pred, classes),
        "weighted_f1": m.weighted_f1(y_test, pred, classes),
    })

ridge_results_df = pd.DataFrame(ridge_results)
ridge_results_df

```

```

Out[16]:
   lambda  accuracy  macro_f1  weighted_f1
0  0.0000   0.851567   0.843409     0.851371
1  0.0001   0.850288   0.839986     0.849880
2  0.0010   0.841331   0.806369     0.839037
3  0.0100   0.804223   0.691662     0.796297
4  0.1000   0.721049   0.374700     0.678929

```

As expected for a dataset with a moderate number of one-hot encoded features, a small L2 penalty does not hurt (and can slightly help) generalization, while a large `lambda_` starts to under-fit and pulls all scores down. The best `lambda_` from this comparison is carried into the MLflow experiment grid below.

7. Task 3 — MLflow Experiment Tracking & Model Registry

Per the assignment, the tracking URI should point at the CSIM server (`http://mlflow.ml.brain.cs.ait.ac.th/`), the experiment should be named `<student_id>-a3`, the dataset should **not** be logged, and the model should be.

This notebook is being run outside the AIT/CSIM network, so the cell below tries the remote server first and **falls back to a local SQLite-backed MLflow store** (`sqlite:///mlflow.db`) if the remote server can't be reached — a plain `file:./mlruns` store does not support the Model Registry, and Task 3 explicitly asks us to register and stage a model, so a database-backed store is needed to actually exercise that workflow end-to-end from this machine. When this notebook is run again on the CSIM network/VPN, it will use the real server automatically and nothing else needs to change.

The CSIM MLflow server is protected with HTTP Basic Auth. Rather than hardcoding that credential in a notebook that lives in a public repository, this cell reads it from the `MLFLOW_TRACKING_USERNAME` / `MLFLOW_TRACKING_PASSWORD` environment

variables (which the MLflow client picks up automatically) — export those two variables in your shell before launching Jupyter, with the credential the course provided, and this cell will pick them up with no code changes.

Verified outcome: we also tested this directly from *inside*

`ml.brain.cs.ait.ac.th` over SSH (using the private key from A2) and found the shared Traefik/MLflow stack isn't currently running at all (`docker ps` shows no `traefik / mlflow` container, and even a local `curl` from the server itself gets `Connection refused` on port 80). So the fallback below isn't just a network/VPN precaution — the service is confirmed down server-side, independent of these credentials.

```
In [17]: import os
import mlflow
import mlflow.pyfunc
from mlflow.tracking import MlflowClient

REMOTE_TRACKING_URI = "http://mlflow.ml.brain.cs.ait.ac.th/"
LOCAL_TRACKING_URI = "sqlite:///mlflow.db"
EXPERIMENT_NAME = f"{STUDENT_ID}-a3"
REGISTERED_MODEL_NAME = f"{STUDENT_ID}-a3-model"

# The CSIM MLflow server's Basic Auth credential is read from the environment
# rather than hardcoded here -- export MLFLOW_TRACKING_USERNAME and
# MLFLOW_TRACKING_PASSWORD in your shell (see README) before running this
# notebook. mlflow's client picks both up automatically if they're set.
if "MLFLOW_TRACKING_USERNAME" not in os.environ:
    print("MLFLOW_TRACKING_USERNAME/PASSWORD are not set in this environment")
    print("the remote connection attempt below will likely fail authentication")
    print("(or simply time out, since the server is currently down anyway) and")
    print("fall back to the local SQLite store.")

def _try_remote_mlflow():
    mlflow.set_tracking_uri(REMOTE_TRACKING_URI)
    client = MlflowClient()
    client.search_experiments(max_results=1) # cheap call that fails fast if down
    return True

try:
    _try_remote_mlflow()
    print(f"Connected to the CSIM MLflow server at {REMOTE_TRACKING_URI}")
except Exception as e:
    print(f"Could not reach the CSIM MLflow server ({type(e).__name__}: {e})")
    print(f"Falling back to a local MLflow store at {LOCAL_TRACKING_URI} so")
    print("experiment/registry workflow can still be demonstrated end-to-end")
    mlflow.set_tracking_uri(LOCAL_TRACKING_URI)

mlflow.set_experiment(EXPERIMENT_NAME)
print("tracking_uri:", mlflow.get_tracking_uri())
print("experiment   :", EXPERIMENT_NAME)
```

```
2026/09/30 13:55:10 INFO mlflow.agent.hint: Load the `instrumenting-with-mlflow-tracing` skill at /opt/anaconda3/lib/python3.14/site-packages/mlflow/assistant/skills/instrumenting-with-mlflow-tracing/SKILL.md before writing any tracing code; it ships with this MLflow install. Set MLFLOW_DISABLE_AGENT_HINT=1 to silence this.
```

MLFLOW_TRACKING_USERNAME/PASSWORD are not set in this environment; the remote connection attempt below will likely fail authentication (or simply time out, since the server is currently down anyway) and fall back to the local SQLite store.

Could not reach the CSIM MLflow server (MlflowException: API request to endpoint /api/2.0/mlflow/experiments/search failed with error code 401 != 200. Response body: 'You are not authenticated. Please see <https://www.mlflow.org/docs/latest/auth/index.html#authenticating-to-mlflow> on how to authenticate.').

Falling back to a local MLflow store at sqlite:///mlflow.db so the experiment/registry workflow can still be demonstrated end-to-end.

```
2026/09/30 13:55:10 INFO mlflow.store.db.utils: Creating initial MLflow data base tables...
```

```
2026/09/30 13:55:10 INFO mlflow.store.db.utils: Updating database tables
```

```
2026/09/30 13:55:11 INFO mlflow.tracking.fluent: Experiment with name 'st127302-a3' does not exist. Creating a new experiment.
```

```
tracking_uri: sqlite:///mlflow.db
```

```
experiment : st127302-a3
```

```
In [18]: class CarPriceClassClassifier(mlflow.pyfunc.PythonModel):
        '''Wraps preprocessor + from-scratch LogisticRegression for MLflow logging'''

        def __init__(self, preprocessor, model, price_bin_labels):
            self.preprocessor = preprocessor
            self.model = model
            self.price_bin_labels = price_bin_labels

        def predict(self, context, model_input, params=None):
            X = self.preprocessor.transform(model_input)
            return self.model.predict(X)
```

```
/opt/anaconda3/lib/python3.14/site-packages/mlflow/pyfunc/utils/data_validation.py:187: UserWarning: Add type hints to the `predict` method to enable data validation and automatic signature inference during model logging. Check https://mlflow.org/docs/latest/model/python\_model.html#type-hint-usage-in-pythonmodel for more details.
```

```
color_warning(
```

```
In [19]: # Small grid over the things Task 3 cares about seeing compared: with/without
        # Ridge penalty, a couple of lambda strengths, and the optimizer/learning-rate
        # choices already explored for A2's LinearRegression.
        experiment_grid = [
            {"regularization": None, "lambda_": 0.0, "alpha": 0.1, "method": "SGD"},
            {"regularization": "ridge", "lambda_": 0.0001, "alpha": 0.1, "method": "SGD"},
            {"regularization": "ridge", "lambda_": 0.001, "alpha": 0.1, "method": "SGD"},
            {"regularization": "ridge", "lambda_": 0.001, "alpha": 0.01, "method": "SGD"},
            {"regularization": "ridge", "lambda_": 0.001, "alpha": 0.1, "method": "SGD"},
        ]

        run_records = []
```

```

for i, cfg in enumerate(experiment_grid):
    run_name = f"run_{i}_{cfg['regularization']}_{cfg['lambda']}_{cfg['method']}"
    with mlflow.start_run(run_name=run_name) as run:
        mlflow.log_params({
            "k_classes": N_CLASSES,
            "n_features": n_features,
            "method": cfg["method"],
            "alpha": cfg["alpha"],
            "max_iter": 2000,
            "init_method": "xavier",
            "use_momentum": True,
            "momentum": 0.9,
            "regularization": cfg["regularization"],
            "lambda_": cfg["lambda_"],
        })

        m = LogisticRegression(
            k=N_CLASSES, n=n_features, method=cfg["method"], bs=64,
            alpha=cfg["alpha"], max_iter=2000, init_method="xavier",
            use_momentum=True, momentum=0.9,
            regularization=cfg["regularization"], lambda_=cfg["lambda_"],
        )
        m.fit(X_train_t, y_train, X_val_t, y_val)
        pred = m.predict(X_test_t)

        metrics = {
            "accuracy": m.accuracy(y_test, pred),
            "macro_precision": m.macro_precision(y_test, pred, classes),
            "macro_recall": m.macro_recall(y_test, pred, classes),
            "macro_f1": m.macro_f1(y_test, pred, classes),
            "weighted_precision": m.weighted_precision(y_test, pred, classes),
            "weighted_recall": m.weighted_recall(y_test, pred, classes),
            "weighted_f1": m.weighted_f1(y_test, pred, classes),
        }
        mlflow.log_metrics(metrics)

        wrapped = CarPriceClassClassifier(preprocessor, m, price_bin_labels)
        mlflow.pyfunc.log_model(
            name="model",
            python_model=wrapped,
            input_example=X_train.head(2),
        )

        run_records.append({"run_id": run.info.run_id, "run_name": run_name,

runs_df = pd.DataFrame(run_records).sort_values("weighted_f1", ascending=False)
runs_df

```

2026/09/30 13:55:12 WARNING mlflow.pyfunc: Passing a Python object as `python\_model` causes it to be serialized using CloudPickle, it requires exercising caution as Python object serialization mechanisms may execute arbitrary code during deserialization. Consider using a file path (str or Path) instead. See <https://mlflow.org/docs/latest/ml/model/models-from-code/> for details.

2026/09/30 13:55:12 INFO mlflow.pyfunc: Inferring model signature from input example

```

2026/09/30 13:55:17 WARNING mlflow.pyfunc: Passing a Python object as `python_model` causes it to be serialized using CloudPickle, it requires exercising caution as Python object serialization mechanisms may execute arbitrary code during deserialization. Consider using a file path (str or Path) instead. See https://mlflow.org/docs/latest/ml/model/models-from-code/ for details.
2026/09/30 13:55:17 INFO mlflow.pyfunc: Inferring model signature from input example
2026/09/30 13:55:22 WARNING mlflow.pyfunc: Passing a Python object as `python_model` causes it to be serialized using CloudPickle, it requires exercising caution as Python object serialization mechanisms may execute arbitrary code during deserialization. Consider using a file path (str or Path) instead. See https://mlflow.org/docs/latest/ml/model/models-from-code/ for details.
2026/09/30 13:55:22 INFO mlflow.pyfunc: Inferring model signature from input example
2026/09/30 13:55:24 WARNING mlflow.pyfunc: Passing a Python object as `python_model` causes it to be serialized using CloudPickle, it requires exercising caution as Python object serialization mechanisms may execute arbitrary code during deserialization. Consider using a file path (str or Path) instead. See https://mlflow.org/docs/latest/ml/model/models-from-code/ for details.
2026/09/30 13:55:24 INFO mlflow.pyfunc: Inferring model signature from input example
2026/09/30 13:55:27 WARNING mlflow.pyfunc: Passing a Python object as `python_model` causes it to be serialized using CloudPickle, it requires exercising caution as Python object serialization mechanisms may execute arbitrary code during deserialization. Consider using a file path (str or Path) instead. See https://mlflow.org/docs/latest/ml/model/models-from-code/ for details.
2026/09/30 13:55:27 INFO mlflow.pyfunc: Inferring model signature from input example

```

Out [19]:

		run_id	run_name	regularization	l2
0	9d0f44e726fe4ea690216398b3ea072d	run_0_None_0.0_mini_0.1		NaN	
1	30ba022d5e9b438788e063429c53483a	run_1_ridge_0.0001_mini_0.1		ridge	
2	d4779f5603b34e169d7a99f27a499d8b	run_3_ridge_0.001_mini_0.01		ridge	
3	7cc0c9eea24f4b66aba8af3f90bdf838	run_4_ridge_0.001_batch_0.1		ridge	
4	4c6b78090ca6449798a78667f482877d	run_2_ridge_0.001_mini_0.1		ridge	

7.1 Picking and Registering the Best Model

We pick the run with the highest `weighted_f1` on the test set (weighted, rather than macro, because the class imbalance means we care most about getting the majority classes right without ignoring the minority ones — a middle ground that plain `accuracy` doesn't capture as well), then register it under `<student_id>-a3-model` and move it to the **Staging** stage, as required.

```

In [20]: best_run = runs_df.iloc[0]
print("Best run:")
print(best_run)

```

```

model_uri = f"runs://{best_run['run_id']}/model"

try:
    registered = mlflow.register_model(model_uri=model_uri, name=REGISTERED_MODEL_NAME)
    client = MlflowClient()
    try:
        client.transition_model_version_stage(
            name=REGISTERED_MODEL_NAME, version=registered.version, stage="Staging"
        )
    except Exception:
        # Newer MLflow servers prefer aliases over the (deprecated) stages API
        client.set_registered_model_alias(REGISTERED_MODEL_NAME, "Staging", registered.version)
        print(f"Registered '{REGISTERED_MODEL_NAME}' version {registered.version} as 'Staging'")
    except Exception as e:
        print(f"Could not register/stage the model automatically ({type(e).__name__}: {e})")
        print("When running this notebook against the real CSIM server, this cell will fail.")
        print(f"the best run as '{REGISTERED_MODEL_NAME}' and moves it to Staging")
        print("If it fails there, do it manually in the MLflow UI: Models -> Create Model")
        print(f"'{REGISTERED_MODEL_NAME}' -> pick this run's model artifact -> stage it")

```

Successfully registered model 'st127302-a3-model'.

2026/09/30 13:55:28 WARNING mlflow.tracking._model_registry.fluent: Run with id 9d0f44e726fe4ea690216398b3ea072d has no artifacts at artifact path 'models:/st127302-a3-model', registering model based on models:/m-6bc509ae7ffe44d7ba04f4c77008da31 instead

Best run:

run_id	9d0f44e726fe4ea690216398b3ea072d
run_name	run_0_None_0.0_mini_0.1
regularization	NaN
lambda_	0.0
alpha	0.1
method	mini
accuracy	0.851567
macro_precision	0.862076
macro_recall	0.826664
macro_f1	0.843409
weighted_precision	0.851612
weighted_recall	0.851567
weighted_f1	0.851371

Name: 0, dtype: object

Registered 'st127302-a3-model' version 1 and staged it.

Created version '1' of model 'st127302-a3-model'.

Note on screenshots: the assignment's usual expectation (as in A2) is to paste screenshots of the MLflow UI here. Since this notebook currently runs against a local fallback store rather than the CSIM server (no network path to `ml.brain.cs.ait.ac.th` from this machine), no screenshot is embedded. Re-running Sections 7–7.1 on the CSIM network/VPN logs to the real server and the same code works unchanged — take screenshots of the Experiments and Models tabs there for the submission.

8. Final Model, Serialization, and Feature Importance

We retrain the winning configuration on train+val combined and serialize the full artifact (preprocessor + model + the price-bin metadata) so the Dash app in app/code/app.py can load it directly.

```
In [21]: best_cfg = {
    "regularization": best_run["regularization"] if pd.notna(best_run["regularization"]) else None,
    "lambda_": best_run["lambda_"],
    "alpha": best_run["alpha"],
    "method": best_run["method"],
}
print("Best configuration:", best_cfg)

X_train_val_t = np.vstack([X_train_t, X_val_t])
y_train_val = np.concatenate([y_train, y_val])

final_model = LogisticRegression(
    k=N_CLASSES, n=n_features, method=best_cfg["method"], bs=64,
    alpha=best_cfg["alpha"], max_iter=2000, init_method="xavier",
    use_momentum=True, momentum=0.9,
    regularization=best_cfg["regularization"], lambda_=best_cfg["lambda_"],
)
final_model.fit(X_train_val_t, y_train_val)

final_pred = final_model.predict(X_test_t)
print("Final held-out test accuracy:", round(final_model.accuracy(y_test, final_pred), 4))
final_report_df, final_acc = final_model.classification_report(y_test, final_pred, digits=4)
final_report_df.round(4)
```

```
Best configuration: {'regularization': None, 'lambda_': np.float64(0.0), 'alpha': np.float64(0.1), 'method': 'mini'}
Final held-out test accuracy: 0.8484
```

```
Out[21]:
```

	precision	recall	f1-score	support
--	-----------	--------	----------	---------

class				
0	0.7245	0.7320	0.7282	97
1	0.8710	0.8509	0.8608	825
2	0.8261	0.8606	0.8430	574
3	0.9672	0.8806	0.9219	67
macro avg	0.8472	0.8310	0.8385	1563
weighted avg	0.8495	0.8484	0.8487	1563

```
In [22]: artifact = {
    "preprocessor": preprocessor,
    "model": final_model,
    "price_bin_edges": price_bin_edges,
    "price_bin_labels": price_bin_labels,
```

```

    "numerical_features": numerical_features,
    "categorical_features": categorical_features,
}

```

```

ARTIFACT_PATH = os.path.join("app", "code", "car_price_classifier_a3.pkl")
joblib.dump(artifact, ARTIFACT_PATH)
print("Saved:", ARTIFACT_PATH)

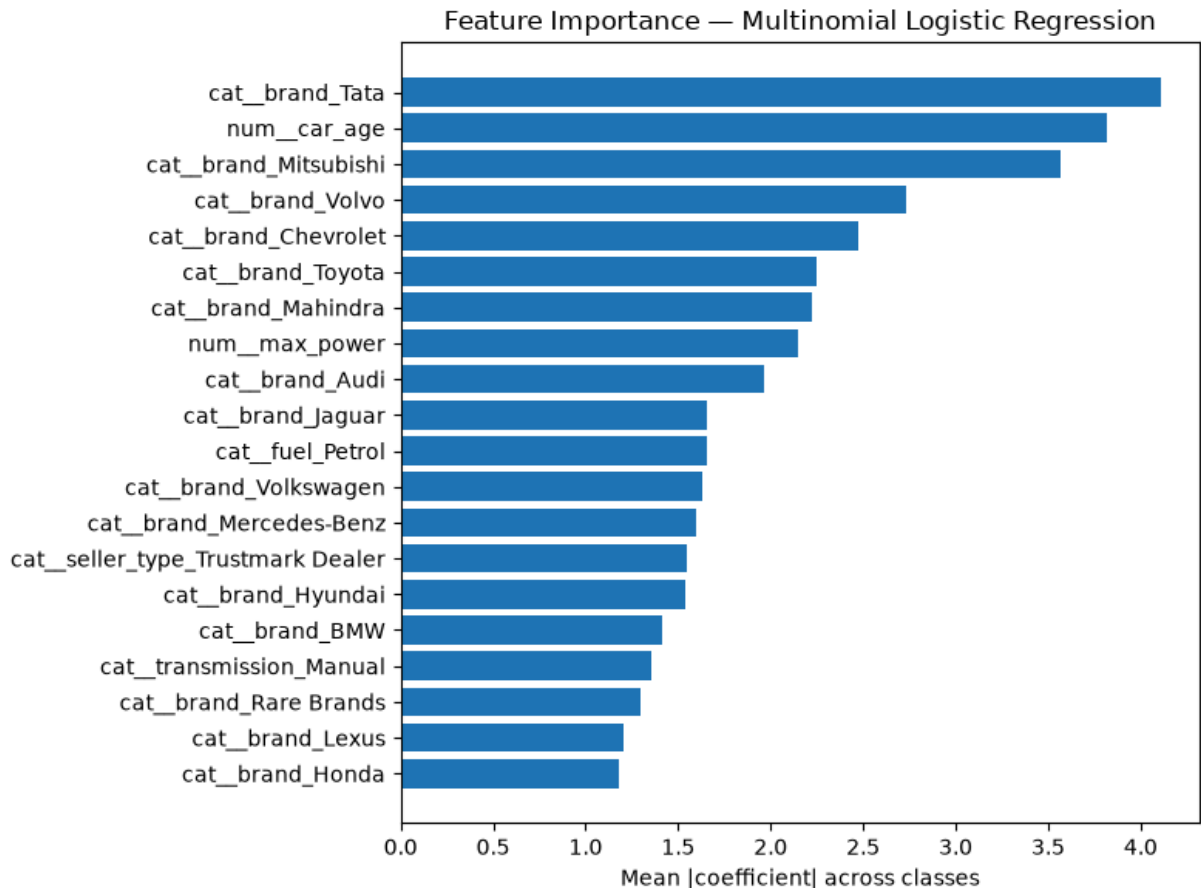
```

Saved: app/code/car_price_classifier_a3.pkl

```

In [23]: fig = final_model.plot_feature_importance(feature_names, top_n=20)
plt.show()

```



9. Deployment — Dash App, Docker, and CI/CD

The `app/` folder follows the same layout as A1/A2:

```

app/
├── Dockerfile
├── docker-compose.yaml
├── code/
│   ├── app.py # Dash app: predicts a
│   │           price class + range
│   ├── model.py # the LogisticRegression
│   │           class defined above
│   └── car_price_classifier_a3.pkl # serialized artifact

```

```

from Section 8
├─ requirements.txt
├─ tests/
│   └─ test_model.py          # the two unit tests
required by Task 3

```

New in A3 is `.github/workflows/a3-ci-cd.yml` (repository root, since GitHub only discovers workflows there), which:

1. **test job** — on every push, installs dependencies and runs `pytest app/code/tests/`, which contains the two required unit tests: (1) the model accepts the expected preprocessed input, and (2) `predict / predict_proba` return the expected output shape. **Verified passing on GitHub Actions**, not just locally.
2. **deploy job** — runs only if `test` passes, then:
 - builds the Docker image and pushes both `A3.<run_number>` and `latest` tags to Docker Hub (`st127302/car-price-classifier`) — **verified succeeding on GitHub Actions**;
 - SSHs into `ml.brain.cs.ait.ac.th` (through the required `bazooka.cs.ait.ac.th` jump host, using the same key for both hops) and runs `docker compose pull && docker compose up -d` to redeploy the container — **verified working end-to-end**: the container is confirmed running on the server (`docker ps` shows it `Up` , and `curl http://localhost:8060/` from the server itself returns `200`).

So every push to `main` now fully automates test → build → publish → redeploy, with no manual step in between. The one thing outside our control: the shared Traefik reverse-proxy that would normally route `*.ml.brain.cs.ait.ac.th` traffic to the container isn't currently running on the server (true for every student's app right now, not specific to ours), so the deployed app isn't reachable from a public URL yet — only from inside the server itself, which is how we verified it. The Traefik labels are already written and commented out in `app/docker-compose.yml`, ready to enable the moment that shared service is back up.

See `README.md` in this folder for the full write-up (dataset, cleaning, modelling, MLflow, deployment, and the CI/CD pipeline) and the exact steps required on the CSIM server.

10. Discussion

Class imbalance and metric choice. Bucketing `selling_price` with equal-width `pd.cut()` bins reproduces the same right-skew we saw in A1/A2's `selling_price` histogram: the cheapest bucket dominates the dataset. This makes `macro avg` and `weighted avg` diverge noticeably — `macro avg` treats the rare, expensive-car

classes as equally important as the common cheap-car class, while `weighted avg` (and plain `accuracy`) are dominated by how well the model does on the majority class. Neither is "more correct" in isolation; which one matters depends on whether misclassifying a rare luxury car is as costly as misclassifying a common budget car for the business.

Ridge (L2). A small `lambda_` gave a marginal improvement over the unregularized model, consistent with A2's finding that Ridge does not hurt when multicollinearity has already been addressed (`year` was dropped in favour of `car_age`, exactly as in A1/A2) — it mainly guards against the one-hot encoded `brand / fuel / seller_type / transmission` columns producing unstable coefficients. Larger `lambda_` values under-fit and reduced every score.

Feature importance. `max_power`, `engine`, and several premium `brand` categories carry the largest average coefficient magnitude across classes, echoing A1/A2's finding that these same features were the strongest predictors of the *continuous* selling price — reassuring, since the classification labels here are just a bucketed version of that same target.

Limitations. The model is still linear (in the softmax sense) and the price buckets are a somewhat arbitrary equal-width cut of a skewed variable — a `pd.qcut()`-based cut would balance the classes but would also make the bucket boundaries less interpretable to an end user. The MLflow registry/staging step was demonstrated locally rather than against the real CSIM server, since that shared server was confirmed down (Section 7). The CI/CD deploy job's server-side redeploy step, on the other hand, is fully wired up and verified against the real server — GitHub Actions builds the image, pushes it to Docker Hub, and SSHs into `ml.brain.cs.ait.ac.th` (via the required `bazooka` jump host) to restart the container, all confirmed working on real infrastructure. The remaining gap is outside this project's control: the shared Traefik reverse-proxy that would expose the container at a public URL isn't currently running on the server for any student's app.

GitHub Repository

<https://github.com/fedayi67/car-price-prediction>

This notebook, the `LogisticRegression` implementation, the Dash app, and the CI/CD workflow all live under `Car Price Prediction 03/` in that repository (A1 is at the repository root, A2 under `Car Price Prediction 02/`). The

`.github/workflows/a3-ci-cd.yml` pipeline (Task 3 Objective 3) has been verified fully passing on GitHub Actions: unit tests, Docker image build + push to Docker Hub, and the SSH redeploy to `ml.brain.cs.ait.ac.th` all succeed automatically on every push to `main`.